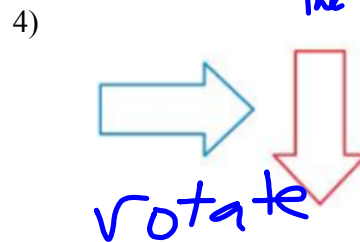
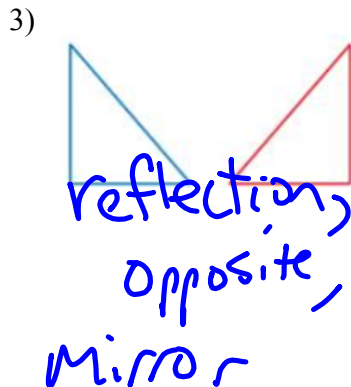
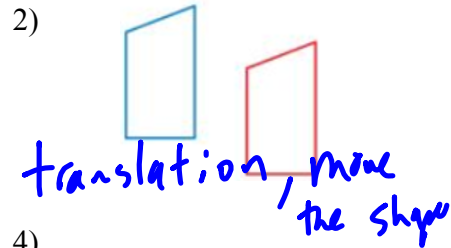
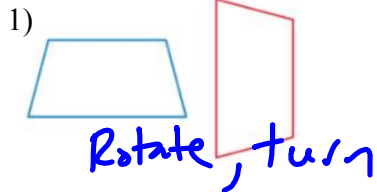


Students will identify transformations, state which ones are rigid motions, and use properties of rigid motions to prove congruence; they will also describe and perform translations.

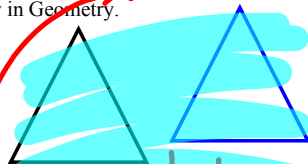
### Do Now

For each diagram, describe how the figure on the left changes into the figure on the right. Be as specific as possible.

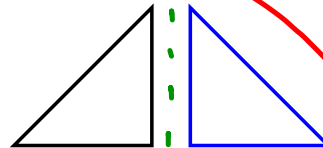


### Transformations

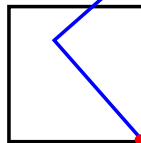
A transformation is a change in the location or size of a figure. Below are the four transformations we will study in Geometry.



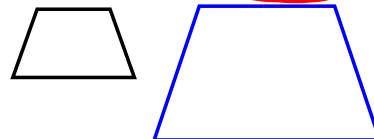
Name: Translation  
Description: Shift up/down, left/right



Name: Reflection  
Description: Flip across a line of symmetry



Name: Rotation  
Description: Turn around a point.



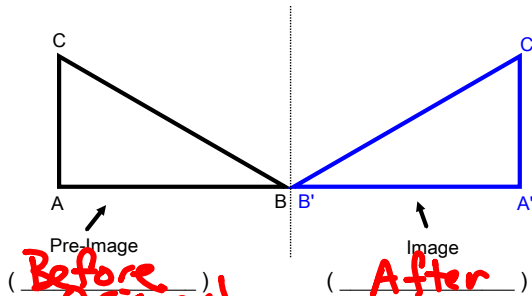
Name: Dilation  
Description: Make bigger (or smaller)

**Rigid Motions**

## Rigid Motions

A Rigid motion is a special type of transformation where the lengths of the sides and the measures of the angles remain the same. 'prim'

Ex 1: A transformation is performed to  $\triangle ABC$  to map it onto  $\triangle A'B'C'$ .



a. What type of transformation has been performed?

Reflection

b. Is this an example of a rigid motion? How do you know?

Yes - sides and  $\angle$ s are unchanged!

## Congruent Shapes

same position

Two shapes are said to be congruent if all corresponding sides are congruent and all corresponding angles are congruent.

Two shapes will always be congruent if there is a set of rigid motions that can be used to map one shape onto the other.

Ex 2: Which figure below shows congruent shapes? Justify your response.

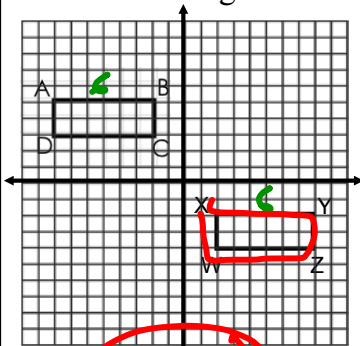


Figure 1

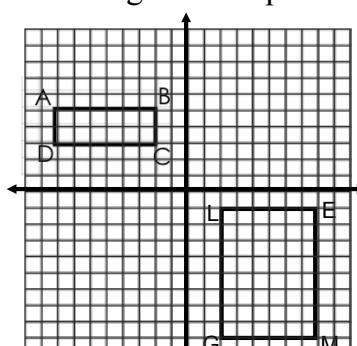
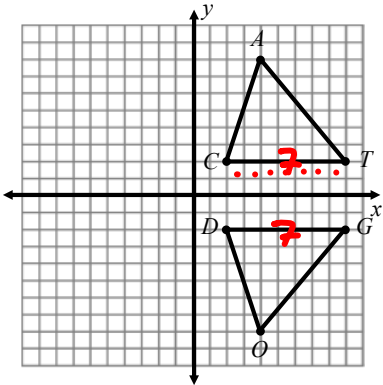


Figure 2

Because all corresponding sides and  $\angle$ s are  $\cong$ .

### Identifying Types of Transformations

Ex 3: a. What type of transformation would map  $\triangle CAT$  onto  $\triangle DOG$ ?



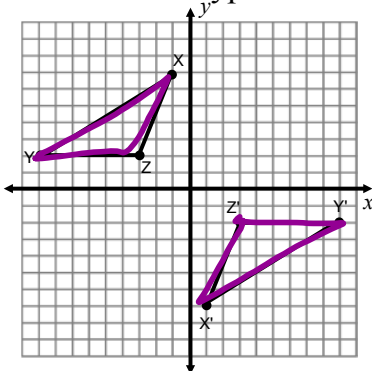
A reflection!

b. Is  $\triangle CAT \cong \triangle DOG$ ? Explain.

"A Reflection is a rigid motion and rigid motions preserve side lengths and angle measures."

You need to memorize this sentence!

Ex 4: a. What type of transformation that would map  $\triangle XYZ$  onto  $\triangle X'Y'Z'$ ?

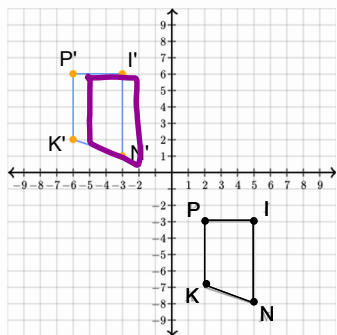


• Rotation

b. Explain why  $\triangle XYZ$  is congruent to  $\triangle X'Y'Z'$ .

A rotation is a rigid motion and rigid motions preserve side lengths +  $\angle$  measures.

Ex 5: a. What type of transformation that maps  $PINK$  onto  $P'T'N'K'$ ?



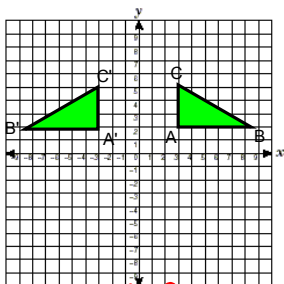
Translation

b. Explain why  $PINK$  is congruent to  $P'T'N'K'$  using the properties of rigid motions.

A translation is a rigid motion, and rigid motions preserve side lengths +  $\angle$  measures.

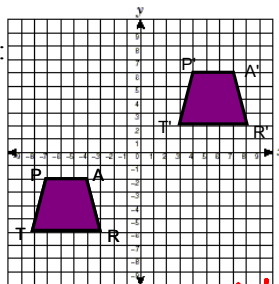
### Practice

Ex 6:



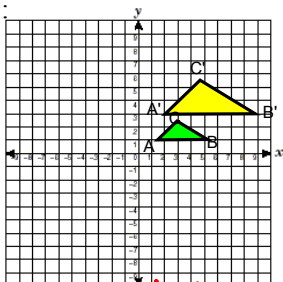
Transformation: Reflection  
Is this a rigid motion? Yes

Ex 7:



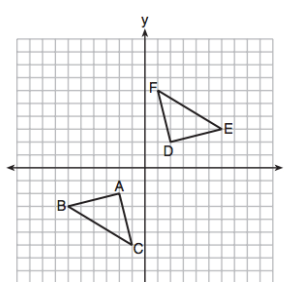
Transformation: Translation  
Is this a rigid motion? Yes

Ex 8:



Transformation: Dilation  
Is this a rigid motion? No

Ex 9:



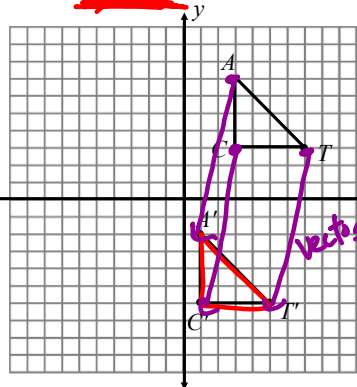
Transformation: Rotation  
Is this a rigid motion? Yes

### Translations

A translation is when you Shift a point, line, or shape **without rotating or flipping it**.

Descriptions of a translations must have vertical  $\updownarrow$  distance and a horizontal  $\leftrightarrow$  distance.

Ex 10: Describe a transformation that would map  $\triangle CAT$  onto  $\triangle C'A'T'$



a. using words.

Move cat 2 units left and 4 units down.

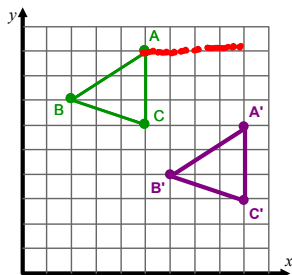
b. using a translation rule.

$$(x, y) \rightarrow (x - 2, y - 4)$$

c. using vector notation.

$\langle -2, -4 \rangle$   
Horizontal, vertical

Ex 11: Describe a transformation that would map  $\triangle ABC$  onto  $\triangle A'B'C'$



a. using words.

Shifts right 4 and down 3

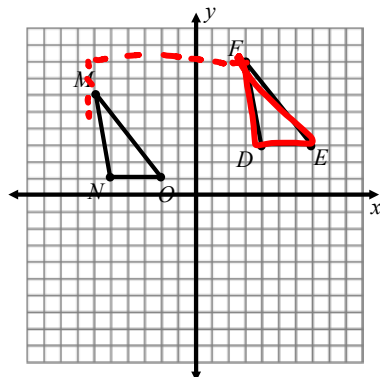
b. using a translation rule.

$$(x, y) \rightarrow (x + 4, y - 3)$$

c. using vector notation.

$\langle 4, -3 \rangle$

Ex 12: Describe a transformation that would map  $\triangle FDE$  onto  $\triangle MNO$ .

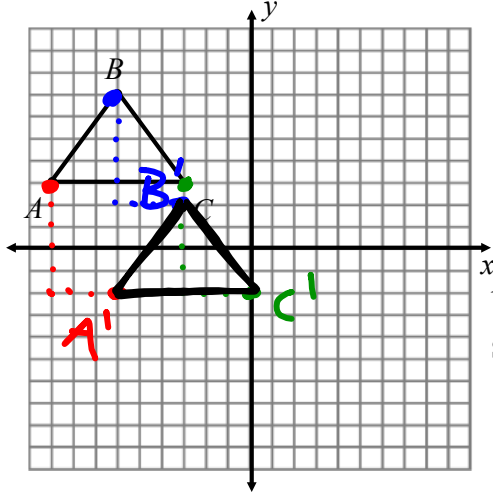


$$(x, y) \rightarrow (x - 9, y - 2)$$

## Performing Translations

Ex 13:  $\triangle ABC$  is translated 5 units down and 3 units to the right. (This can be written using symbols as:  $(x+3, y-5)$ ).

a. What are the coordinates of  $\triangle A'B'C'$ ?



$$A(-9,3) \xrightarrow{+3, -5} A'(-6, -2)$$

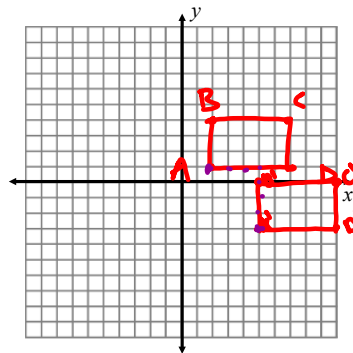
$$B(-6,7) \xrightarrow{+3, -5} B'(-3, 2)$$

$$C(-3,3) \xrightarrow{+3, -5} C'(0, -2)$$

b. Plot the image  $\triangle A'B'C'$  on the graph shown.

Ex 14: Given quadrilateral  $ABCD$  with  $A(2,1)$ ,  $B(2,4)$ ,  $C(7,4)$  and  $D(7,1)$ . A translation of  $(x,y) \rightarrow (x+3, y-4)$  is applied to  $ABCD$ .

a. What are the coordinates of  $A'B'C'D'$ ?



$$A(2,1) \xrightarrow{+3, -4} A'(5, -3)$$

$$B(2,4) \xrightarrow{+3, -4} B'(5, 0)$$

$$C(7,4) \xrightarrow{+3, -4} C'(10, 0)$$

$$D(7,1) \xrightarrow{+3, -4} D'(10, -3)$$

b. Explain why  $ABCD$  and  $A'B'C'D'$  are congruent.

A translation is a rigid motion and rigid motions preserve side lengths and  $\angle$  measures.

## HW Diagram

